

Chapter 3 Notes

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1. **Theorem 3.14.** Let (s_n) be monotonic. Then (s_n) converges if and only if it is bounded.

2. **Definition 3.16.** Let $(s_n) \in \mathbb{R}$ and (s_{n_i}) be a subsequence of (s_n) . Define

$$E = \{x \in \mathbb{R} \cup \{-\infty, \infty\} \mid (s_{n_i}) \rightarrow x\}.$$

Then $\sup E = \limsup_{n \rightarrow \infty} s_n$ and $\inf E = \liminf_{n \rightarrow \infty} s_n$ are said to be the *upper* and *lower* limits of (s_n) .

3. **Theorem 3.19.** If $(s_n) \leq (t_n)$, then $\liminf_{n \rightarrow \infty} s_n \leq \liminf_{n \rightarrow \infty} t_n$ and $\limsup_{n \rightarrow \infty} s_n \leq \limsup_{n \rightarrow \infty} t_n$.

4. **Theorem 3.20.**

- (a) $p > 0 \implies \lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.
- (b) $p > 0 \implies \lim_{n \rightarrow \infty} \sqrt[p]{p} = 1$.
- (c) $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.
- (d) $\forall \alpha \in \mathbb{R} \ p > 0 \implies \lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n} = 0$.
- (e) $|x| < 1 \implies \lim_{n \rightarrow \infty} x^n = 0$.

5. **Theorem 3.22.** $\sum a_n$ converges if and only if

$$(\forall \varepsilon > 0)(\exists N \in \mathbb{Z}) \ m \geq n \geq N \implies \left| \sum_{i=n}^m a_i \right| \leq \varepsilon.$$

6. **Theorem 3.23.** If $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

7. **Theorem 3.24.** A series of non-negative terms converges if and only if its partial sums form a bounded sequence.

8. **Theorem 3.25.**

- (a) If $|a_n| \leq c_n$ and $\sum c_n$ converges, then $\sum a_n$ converges.
- (b) If $a_n \geq d_n \geq 0$ and $\sum d_n$ diverges, then $\sum a_n$ diverges.

9. **Theorem 3.26.** If $0 \leq x < 1$, then $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$; furthermore, if $x \geq 1$, then $\sum_{n=0}^{\infty} x^n$ diverges.
10. **Theorem 3.27.** Let (a_n) be monotonically decreasing and nonnegative. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if
- $$\sum_{i=0}^{\infty} 2^i a_{2^i} = a_1 + 2a_2 + 4a_4 + \cdots$$
- converges.
11. **Theorem 3.28.** If $p > 1$, then $\sum \frac{1}{n^p}$ converges; furthermore, if $p \leq 1$, then $\sum \frac{1}{n^p}$ diverges.
12. **Theorem 3.31.** $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$.
13. **Theorem 3.33.** For $\sum a_n$ define $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$. Then
- (a) If $\alpha < 1$, then $\sum a_n$ converges;
 - (b) If $\alpha > 1$, then $\sum a_n$ diverges;
 - (c) If $\alpha = 1$, then the test is inconclusive.
14. **Theorem 3.34.**
- (a) $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 \implies \sum a_n$ converges.
 - (b) $\left| \frac{a_{n+1}}{a_n} \right| \geq 1 \implies \sum a_n$ diverges.
15. **Theorem 3.37.** Let $(c_n) \in \mathbb{R}_{>0}$. Then $\liminf_{n \rightarrow \infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n \rightarrow \infty} \sqrt[n]{c_n}$ and $\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}$.
16. **Theorem 3.39.** For $\sum c_n z^n$ define $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{c_n}$ and $R = \frac{1}{\alpha}$. Then if $|z| < R$, then $\sum c_n z^n$ converges; furthermore, if $|z| > R$, then $\sum c_n z^n$ diverges.
17. **Theorem 3.42.** Let the partial sums of A_n of $\sum a_n$ form a bounded sequence, (b_n) be monotonically decreasing, and $\lim_{n \rightarrow \infty} b_n = 0$. Then $\sum a_n b_n$ converges.
18. **Theorem 3.43.** Let $(|c_n|)$ be monotonically decreasing, $c_{2m-1} \geq 0$, $c_{2m} \leq 0$, and $\lim_{n \rightarrow \infty} c_n = 0$. Then $\sum c_n$ converges.
19. **Theorem 3.44.** Let $\sum c_n z^n$ have a radius of convergence of 1, (c_n) be monotonically decreasing, and $\lim_{n \rightarrow \infty} c_n = 0$. Then $\sum c_n z^n$ converges at every point on the circle $|z| = 1$, except possibly at $z = 1$.
20. **Theorem 3.45.** If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.
21. **Theorem 3.47.** Let $\sum a_n = A$ and $\sum b_n = B$. Then $\sum (a_n + b_n) = A + B$ and $\sum c a_n = cA$.

22. **Definition 3.48.** For $\sum a_n$ and $\sum b_n$ define

$$c_n = \sum_{i=0}^n a_i b_{n-i}.$$

Then $\sum c_n$ is said to be the *product* of $\sum a_n$ and $\sum b_n$.

23. **Theorem 3.50.** Let $\sum_{n=0}^{\infty} a_n$ converge absolutely, $\sum_{n=0}^{\infty} a_n = A$, $\sum_{n=0}^{\infty} b_n = B$, and $c_n = \sum_{i=0}^{\infty} a_i b_{n-i}$. Then $\sum_{n=0}^{\infty} c_n = AB$.